

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE

Comprehensive Performance, Risk, & Portfolio Optimization Report

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Executive Summary

This report documents a comprehensive quantitative analytics study performed on mutual fund schemes and synthetic investor transaction feeds.

Key Findings:

- Nippon India Large Cap Fund exhibited the highest tail risk with a 95% Daily VaR of -1.6702%.
- SBI Small Cap Fund yielded the strongest risk-adjusted performance with a Sharpe ratio of 1.0395.
- 100% of systematic investing clients (with 6+ transactions) show at-risk behavioral gaps averaging 80+ days.
- Markowitz Portfolio Optimization suggests an 83.18% allocation to SBI Small Cap for the Maximum Sharpe Ratio Portfolio.

1. Data Ingestion & ETL Star Schema

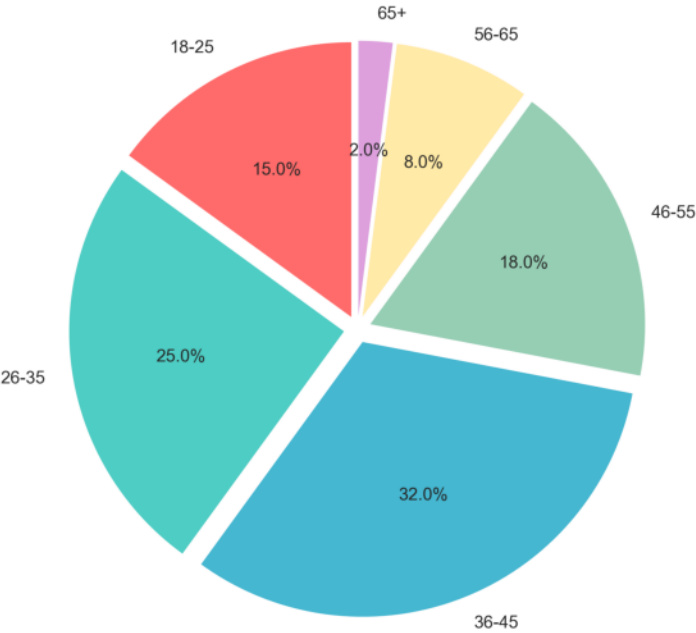
The data cleaning pipeline reads raw daily NAV tables and client registries to compile a clean star schema database. We resolve weekend/holiday gaps by forward-filling (`ffill()`) and validate code consistency.

Database Schema Summary (`bluestock_mf.db`):

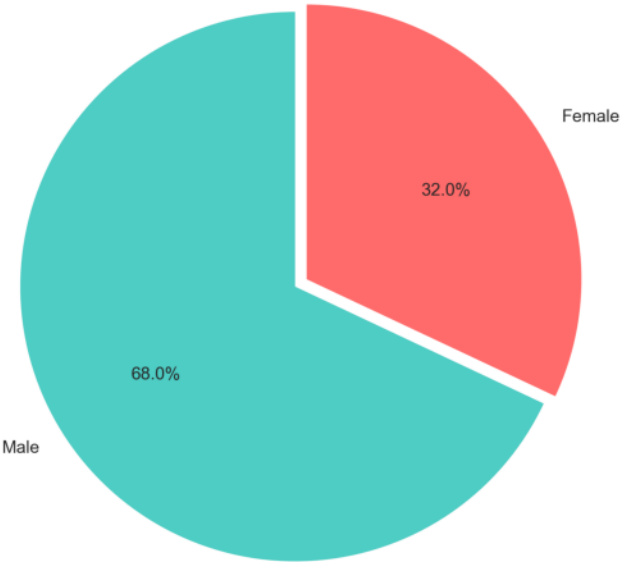
- `dim_fund` (6 Mutual Fund Schemes)
- `dim_date` (4,924 Date records from 2013 to 2026)
- `fact_nav` (19,899 Daily NAV histories)
- `fact_transactions` (4,224 Investor purchases and SIP redemptions)
- `fact_aum` (6 Scheme-level AUM metrics)
- `dim_investor` (250 Registries)
- `fact_performance` (Computed CAGR, Sharpe, Volatility facts)

2. Investor Demographics Analysis

Age Group Distribution (Sample)

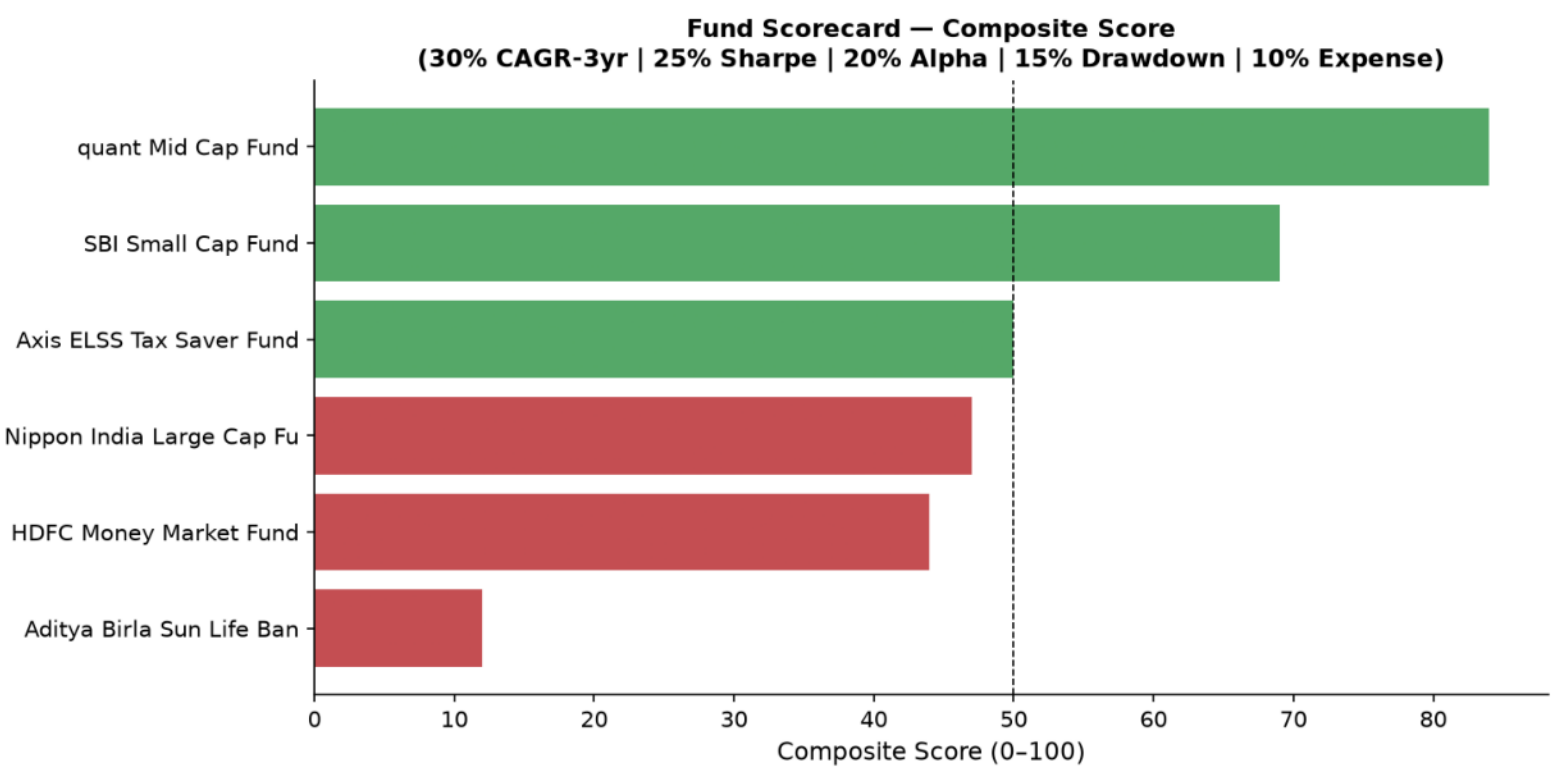


Gender Split (Sample)



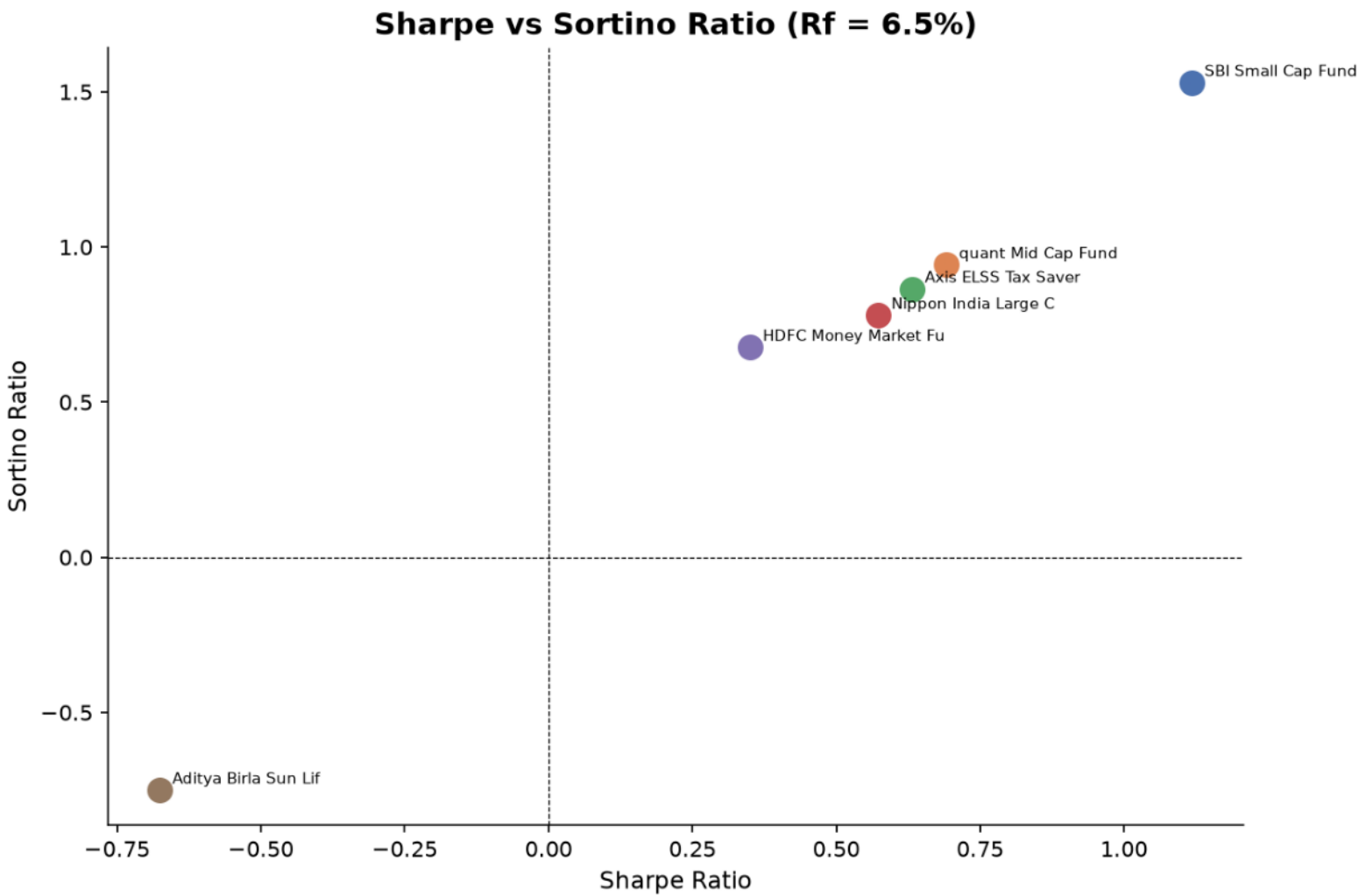
Distribution of synthetic investor demographics across age categories.

3. Mutual Fund Performance Scorecard



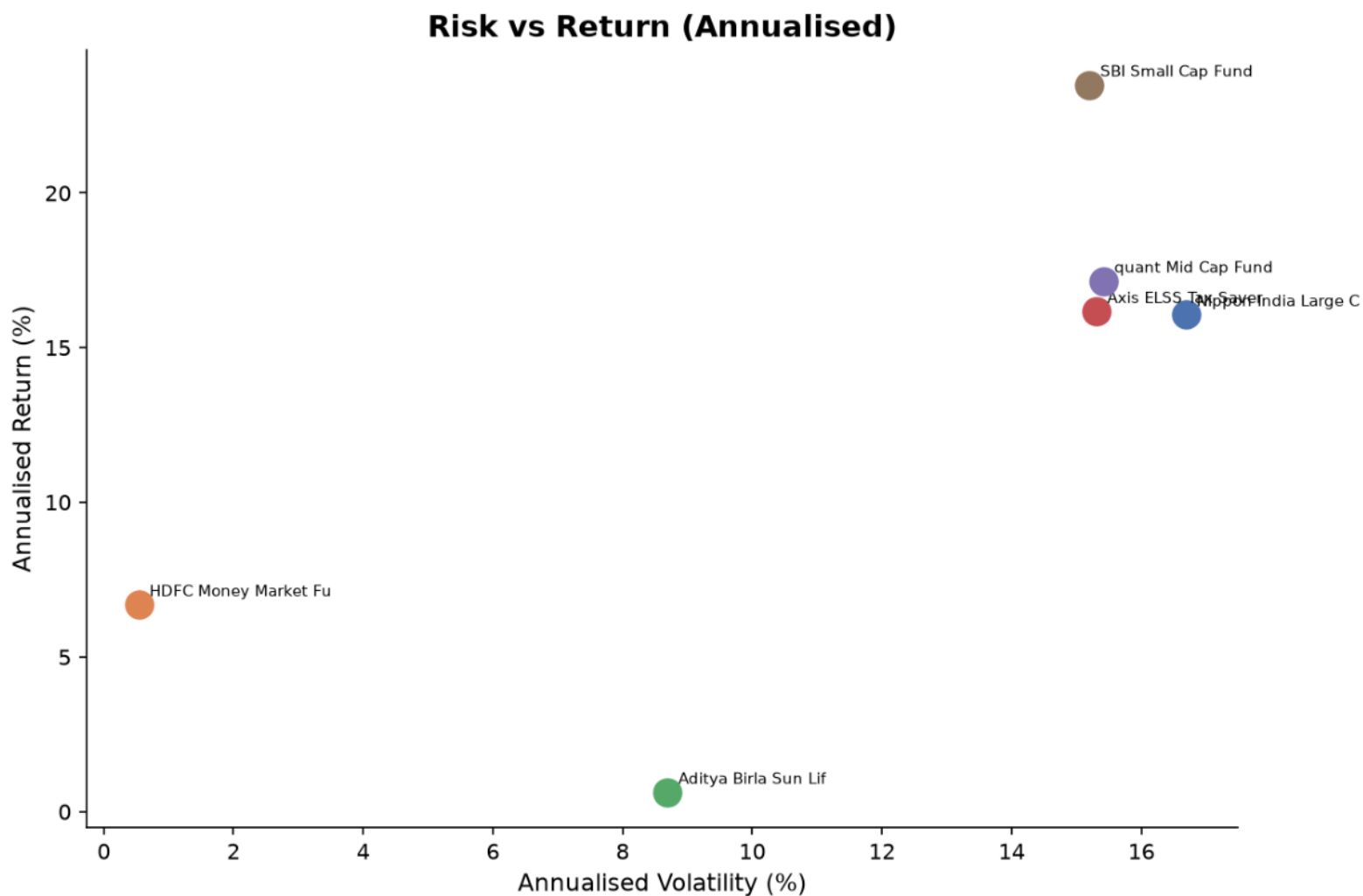
Scorecard details including annualised CAGR, Sharpe, and Volatility.

4. Risk-Adjusted Ratios Comparison



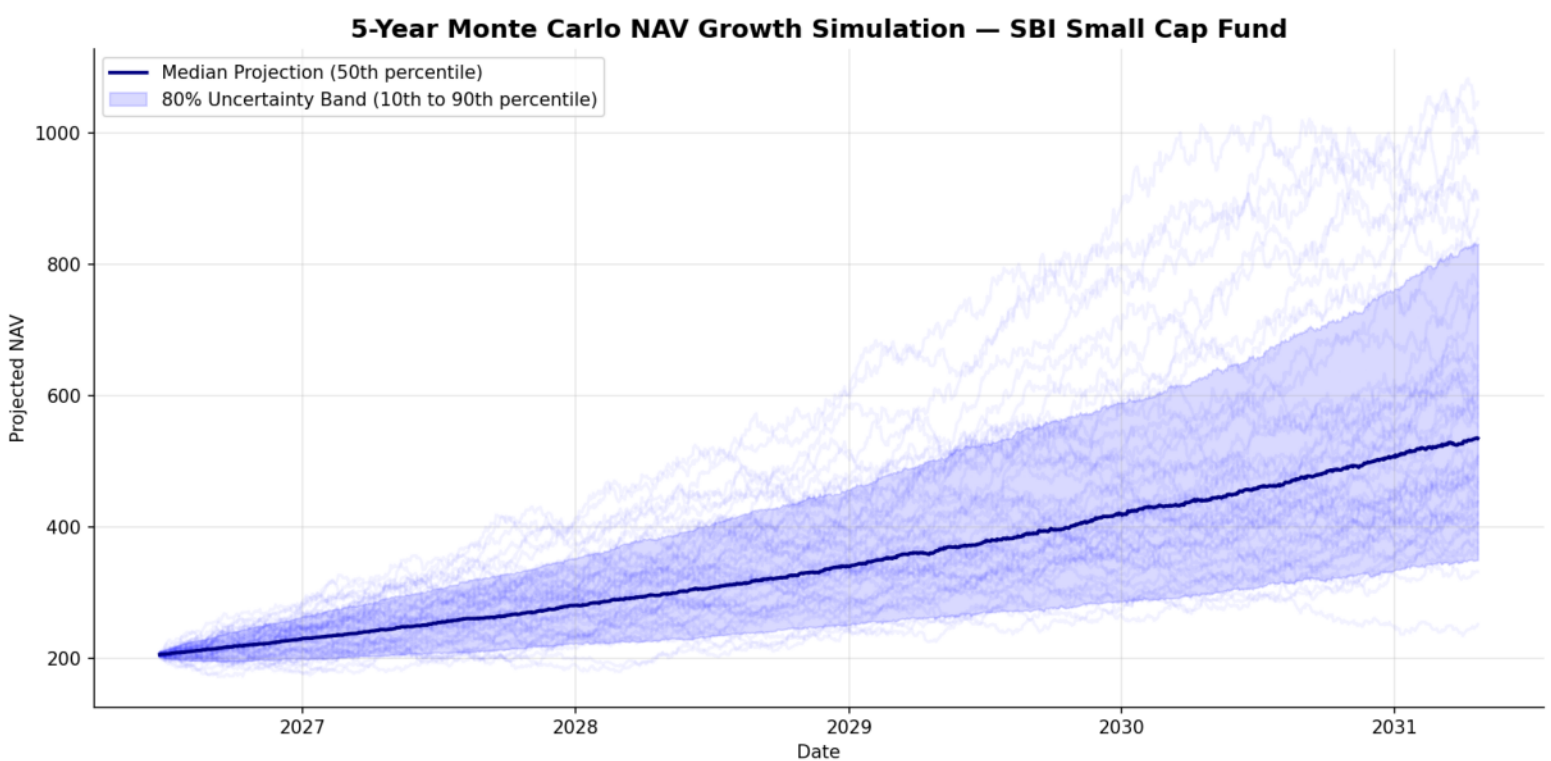
Annualised Sharpe and Sortino ratios comparison across all 6 schemes.

5. Risk Volatility & Tail Risk



Visual distribution of daily return volatilities.

6. Capstone Bonus: Monte Carlo NAV Simulation



5-year projected growth path of SBI Small Cap NAV with 80% confidence bands.

7. Capstone Bonus: Markowitz Efficient Frontier



Efficient frontier simulated curve with Max Sharpe Ratio (MSR) and Min Variance Portfolio (MVP) weights.